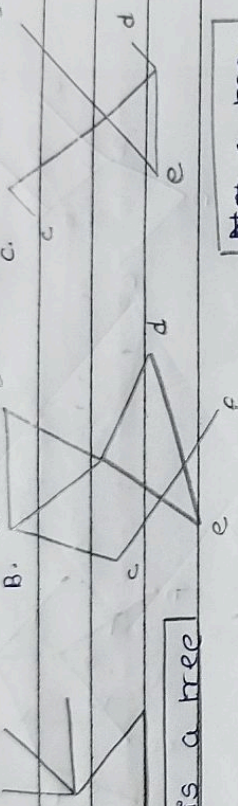


Trees

Q5) (b) (5)

o Trees :- A tree is a connected undirected graph with no simple circuit.

eg:-



It is a tree

Not a tree

Not a tree

Solⁿ → The graph A and C are tree because no circuit is formed.

→ Graph B is not a tree because it is connected & forms a circuit i.e.

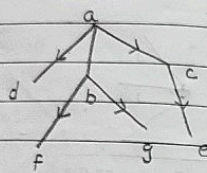
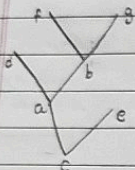
a - d - e - b - a

Theorem: An undirected graph is a tree if and only if there is a unique simple path between any 2 vertices.

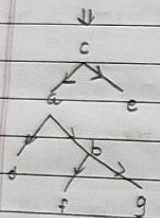
o Rooted Tree: A rooted tree is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

eg:- Draw the rooted trees for the given graph with A & C as its roots

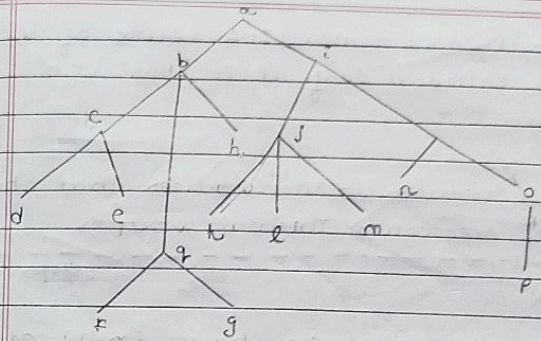
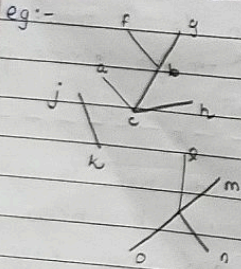
Graph A:



Already a rooted tree



eg:- Forests:- A graph in which every connected component is a tree is called a forest.



Parent vertex: In a rooted tree 'T', u is called parent of v if there is a unique edge from u to v

Child vertex: If u is parent of v then v is child of u.

Siblings: Vertices with the same parent are called siblings

Ancestors: The ancestors of the vertex v other than the root are the vertices in the path from the root to the vertex v

Descendants: The descendants of vertex v are all those vertices that have v as an ancestor

• leaf :- [Pendant vertex] :- A vertex of a tree is called a leaf if it has no children.

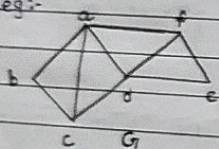
• Internal vertices :- The vertices that have children are called Internal vertices.

9/4/25

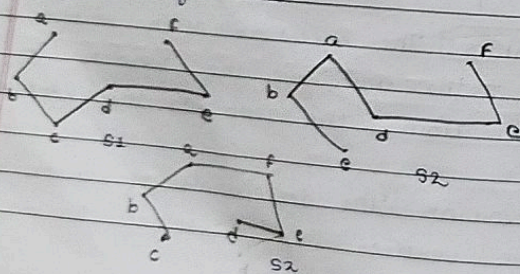
Theorem :- A tree with n vertices have $(n-1)$ edge.

Spanning trees :- Let G be the simple graph. A spanning tree of G is a subgraph of G that is a tree containing every vertex of G .

eg:-



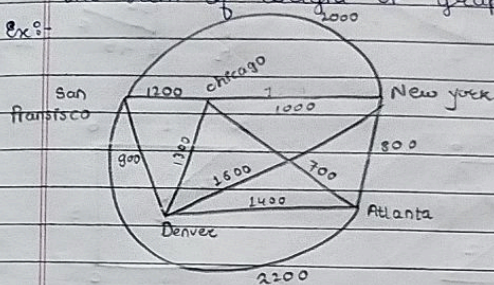
Subgraphs of G are :-



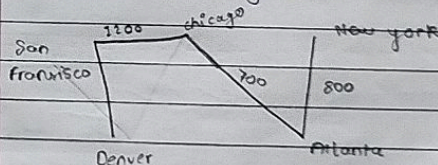
S_1, S_2 and S_3 are spanning graph of G .

• Minimum Spanning tree :- Minimum spanning tree of a graph is a spanning tree which the sum of weight of graph is minimum.

Ex:-

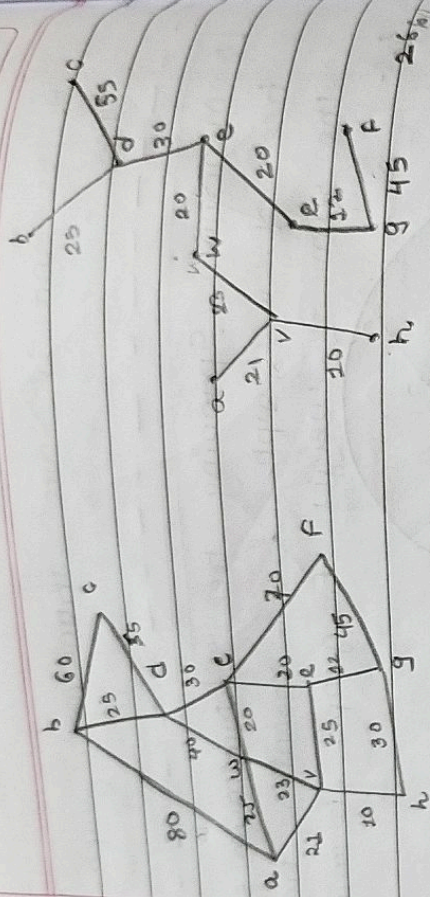


Minimum spanning tree is :-



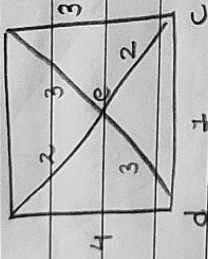
Ans:- Therefore, the sum of weight is $900 + 800 + 1200 + 700 = 3600$.

Q2]

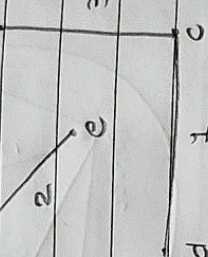


Q2] Drc

a 1 b

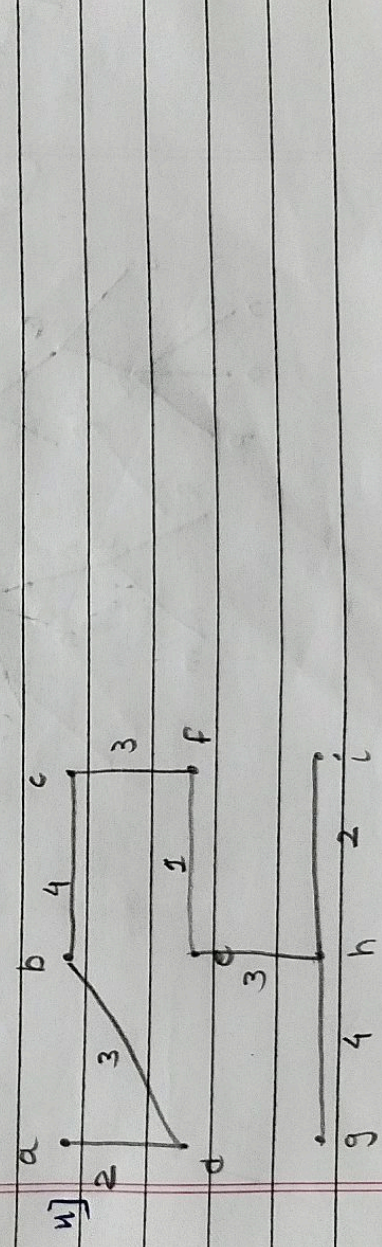
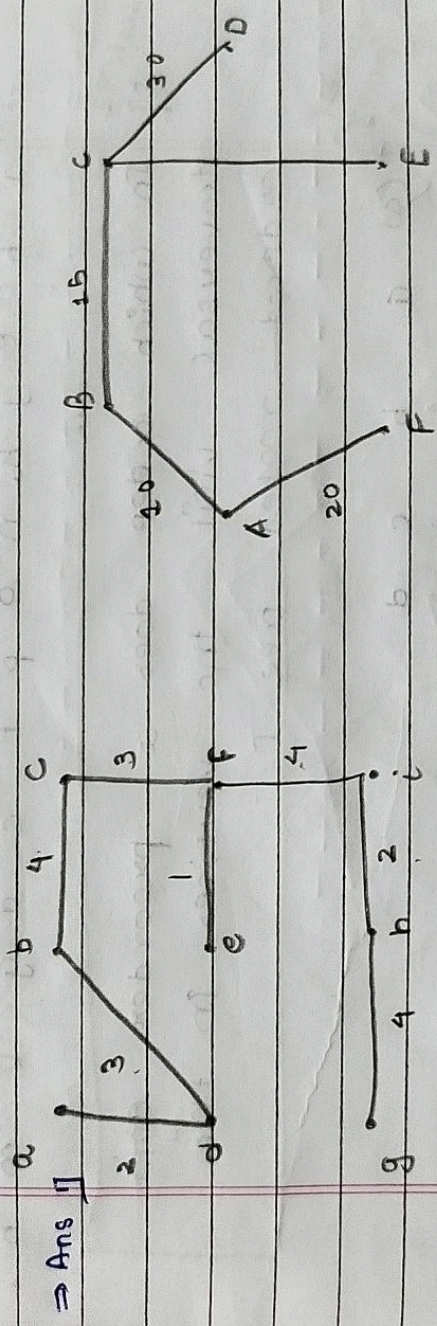
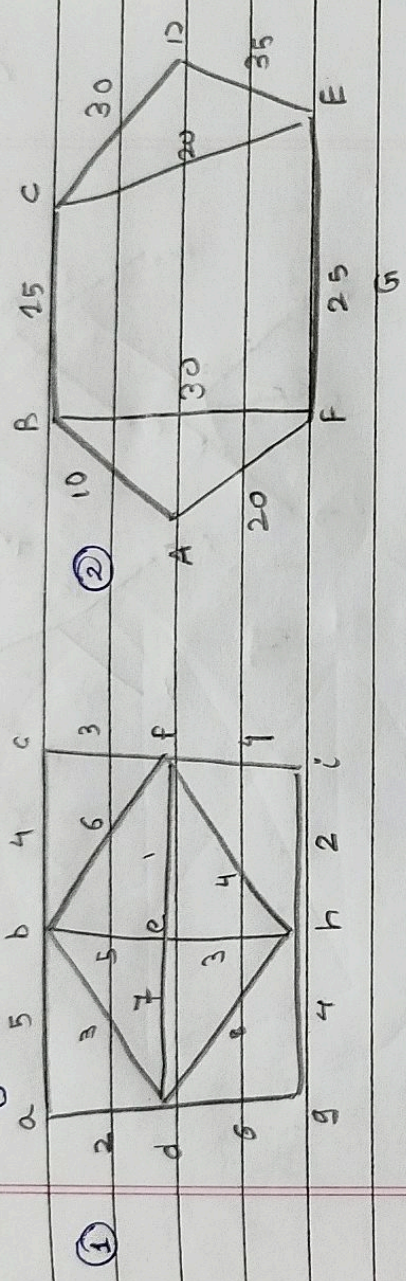


a 1 b



Q3] Sh

Q1 Find the minimum spanning tree for the given weighted graph using Prim's algorithm:-

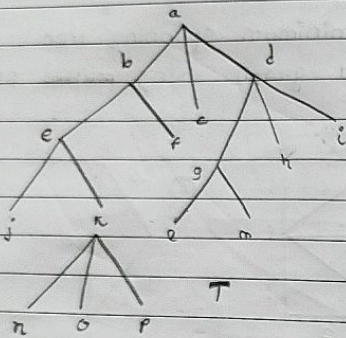


iii] is minimum spanning tree

o Tree Traversal :

1) Preorder Traversal

Answer
[part II]
[Final]

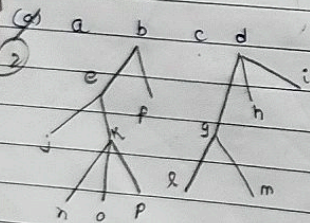


a b e j k n o p f c d g l m h i

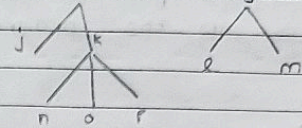
Question

2] In which order does a preorder traversal visit the vertices in the ordered rooted tree T

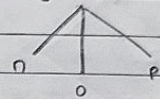
Answer
[part I]
[Starting]



step 2 a b e f c d g h i

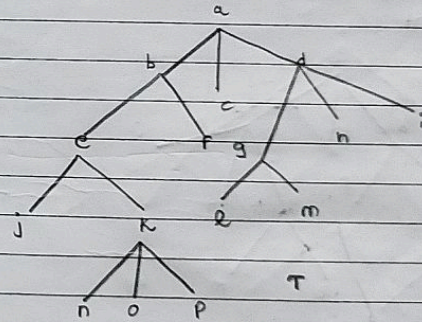


step 3 a b e j k f c d g l m h i

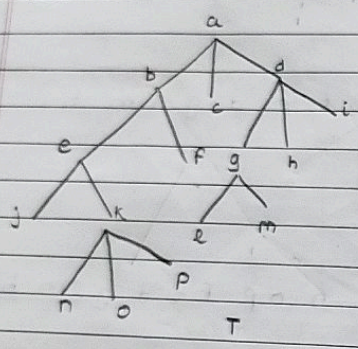


step 4 a b e j k n o p f c d g l m h i

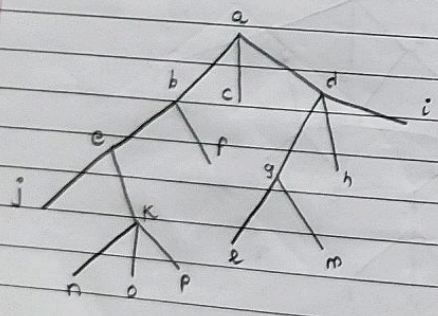
step 5 Visiting the root a, visit sub trees left to right.



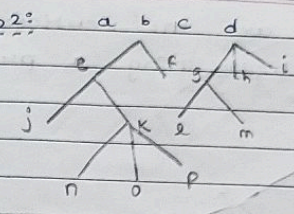
Q1] In which order does a preorder traversal visit the vertices in the ordered rooted tree T.



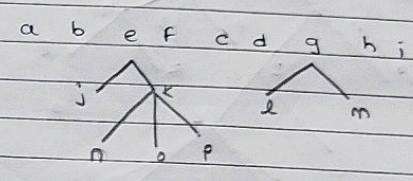
Ans ⇒ Step 1: Visiting the root a, visit sub tree left to right



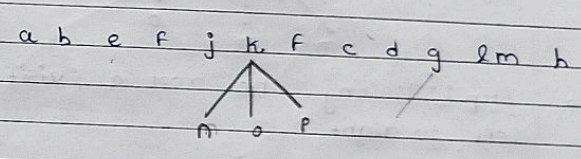
Step 2:



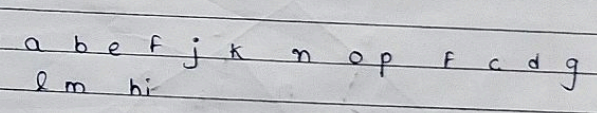
Step 3:



Step 4:

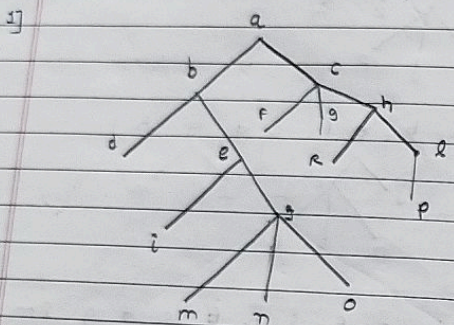


Step 5:

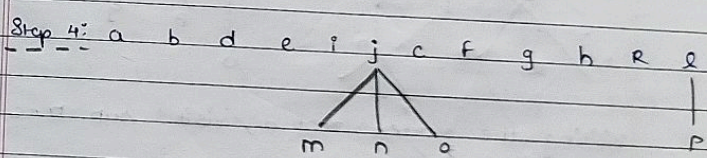
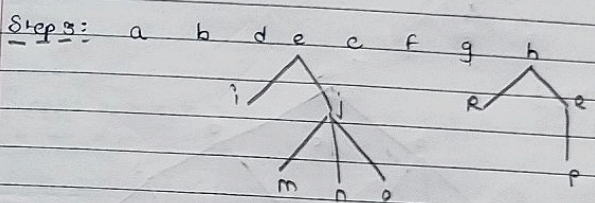
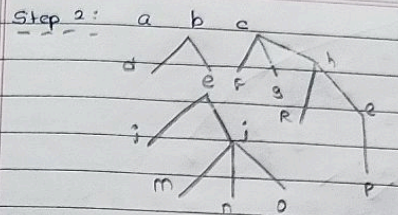
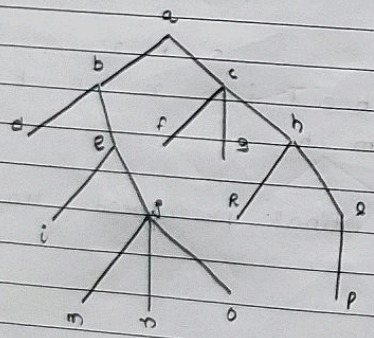


Is the final preordered traversal

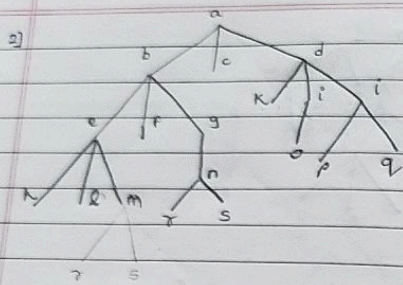
Q2] In which order are the vertices of the ordered rooted trees using a preorder traversal:



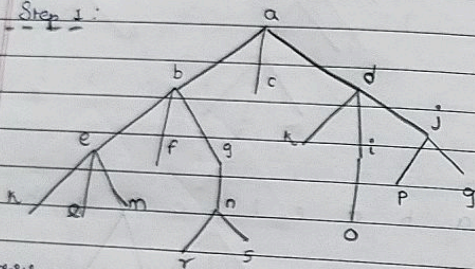
Step 1: Visiting the root a, visit sub tree left to right



Step 5: a b d e i j m n o c f g h r p

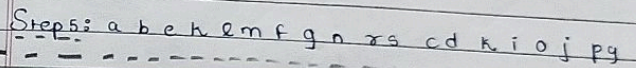
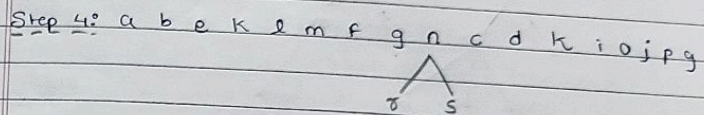
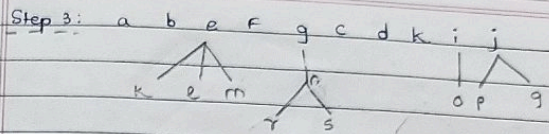
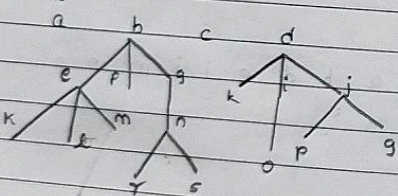


Ans $\Rightarrow$ Step 1:



Visiting the vertex root a , visit sub tree left to right

Step 2:

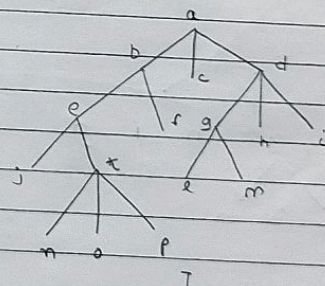


Step 6:

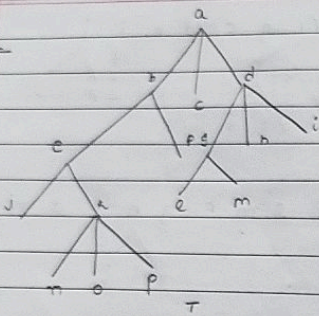
Post order Traversal:-

9] In which order are the vertices of the ordered rooted trees below visited using a post ordered traversal:-

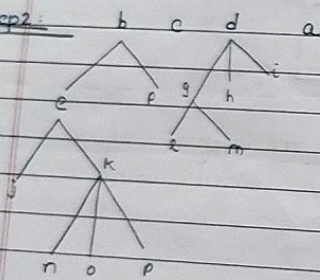
(1)



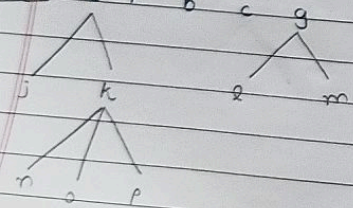
Step 1:



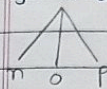
Step 2:



3: e f b c g h i d a



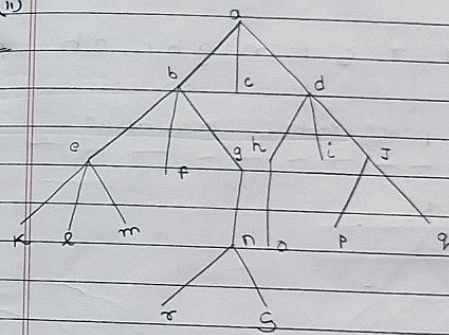
Step 4: j k e f b c g m g h i d a



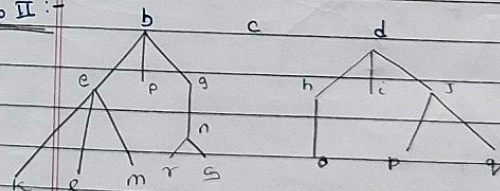
Step 5: j n o p k e f b c g m g h i d a
∴ Is the final post order transversal

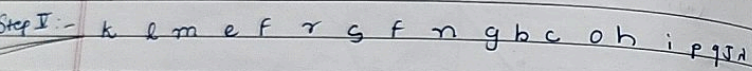
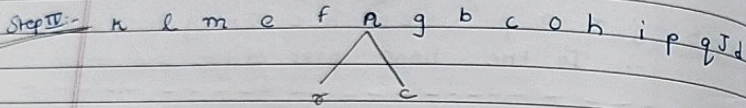
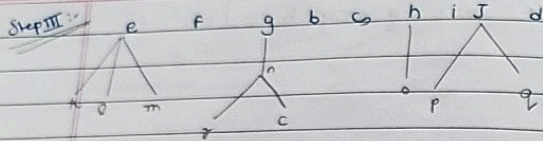
(11)

Step I:



Step II:-

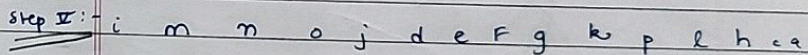
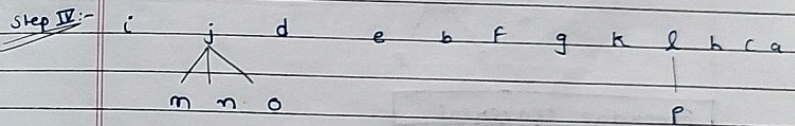
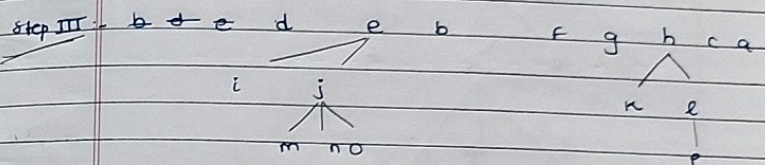
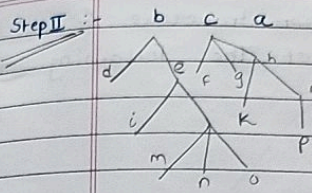
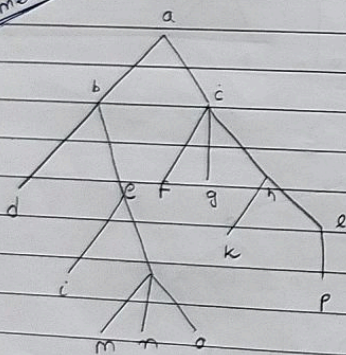




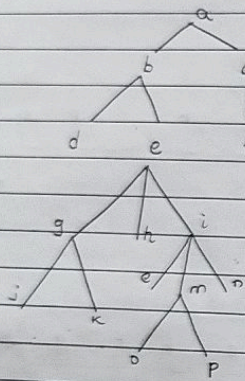
Above is post order traversal

(iii) Homework

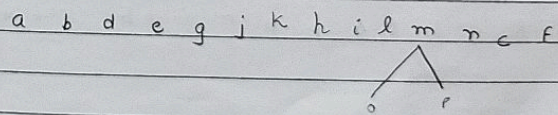
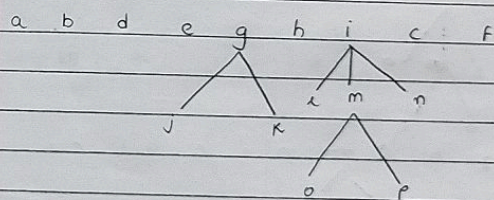
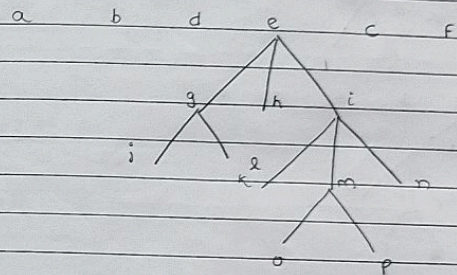
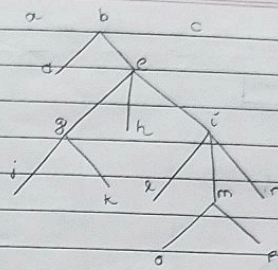
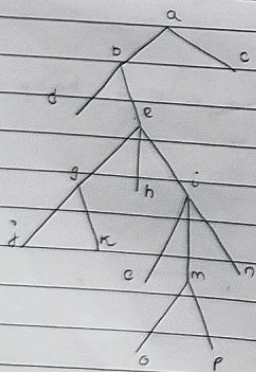
Step I:-



Q1 Find the preorder, inorder, postorder & level order traversal of the following rooted trees.



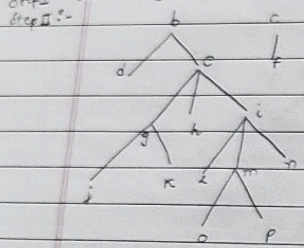
Preorder traversal



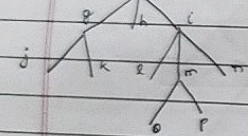
a b d e g j k h i l m o p n c f

POST ORDAL TRAVERSAL

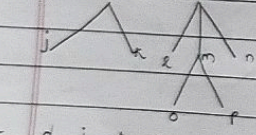
Step I:- Draw diagram of tree



Step II:-



III:-



d j k g h l m n i e b f c a

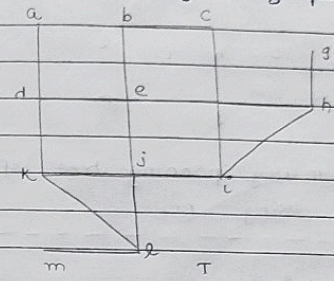


d j k g h l o p m n i e b f c a

Spanning tree $\rightarrow$ tree having all vertices of graph.

Breadth - First Search:-

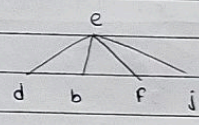
g) Find the spanning tree using Breadth - First Search (BFS) for the given graphs.



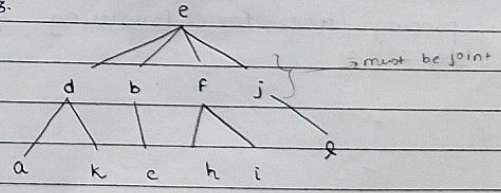
$\Rightarrow$ Step I:- 1.

e

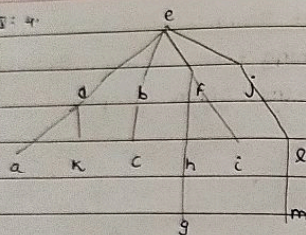
Step II:- 2.



Step III:- 3.

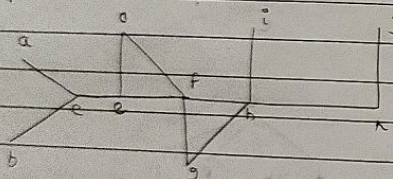


Step II: 4



This is the ~~summary~~ spanning tree of the given graph using BFS.

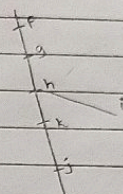
Depth First Search:-



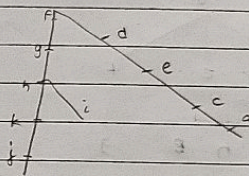
Q1] Use depth first search to produce a spanning tree for following graph
⇒ Step I: 1



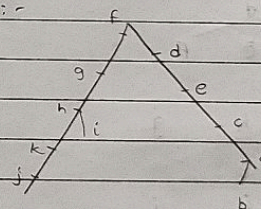
Step II: 2



Step III:-

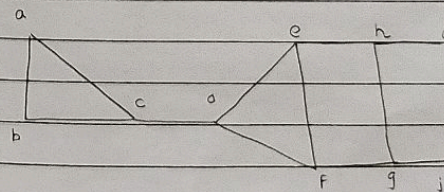


Step IV:-



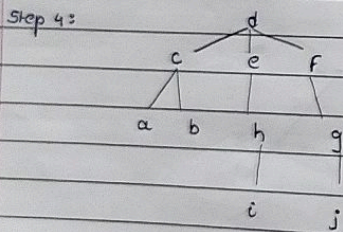
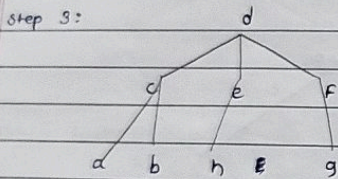
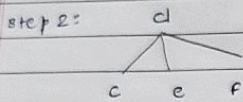
Q2] Find BFS and DFS of the following graphs

(1)

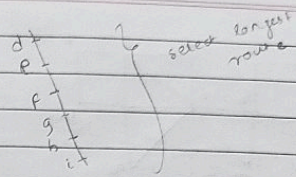


G1 BSF :-

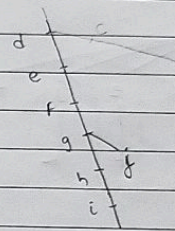
Step 1: d



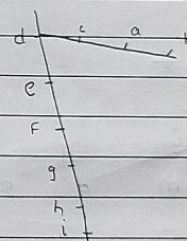
DSF :- Step 1:



Step II:



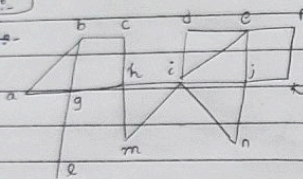
Step III:



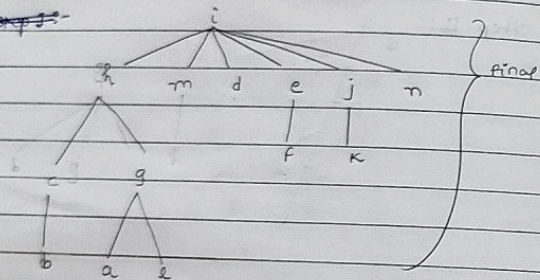
G2

BSF :-

Step I :-



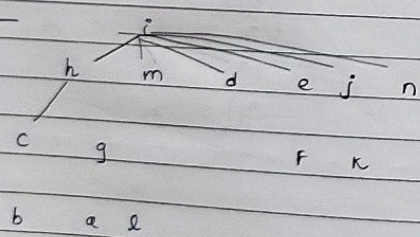
Step II :-



Step III :-

f

Step IV :-

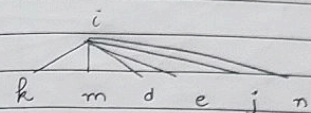


BSF :-

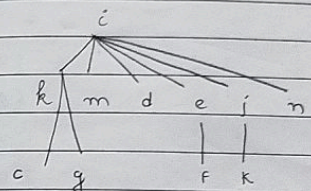
Step I :-

i

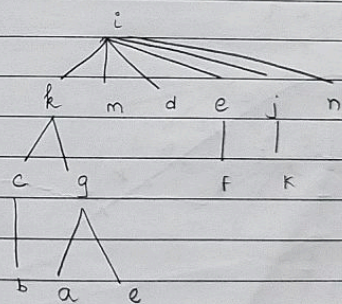
Step II :-



Step III :-



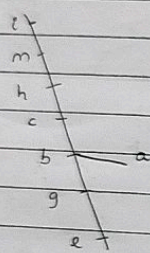
Step IV :-



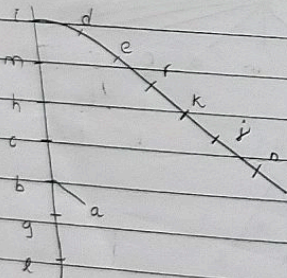
DSF :- Step I :-



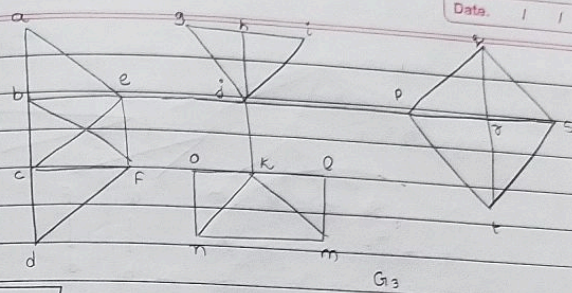
Step II



Step III :-



G₃
(iii)

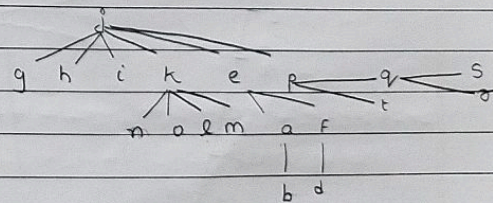
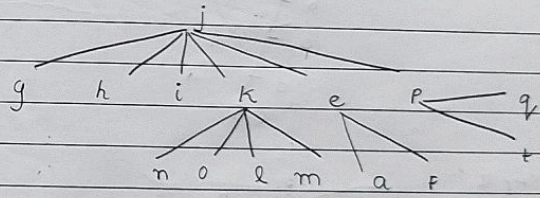
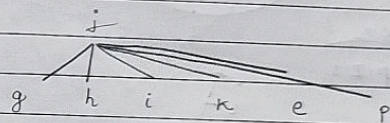


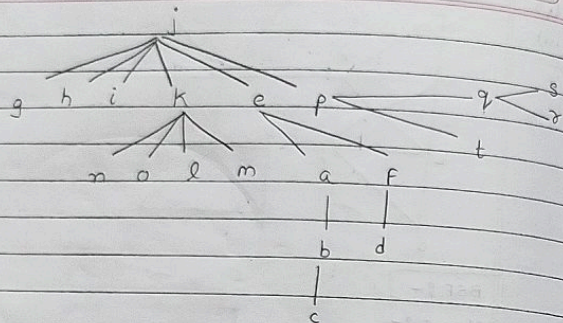
BSF :-

Step I :-



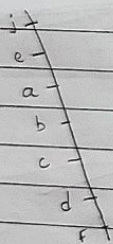
Step II :-



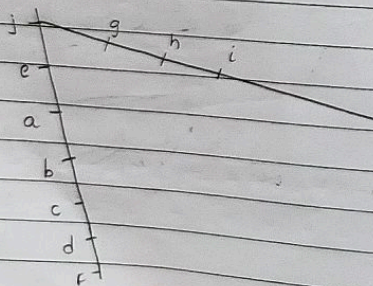


DFS :-

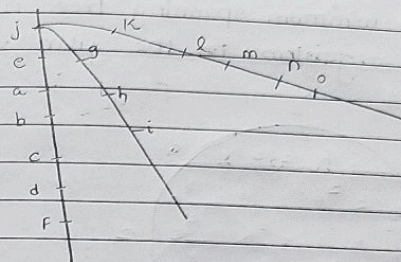
Step I :-



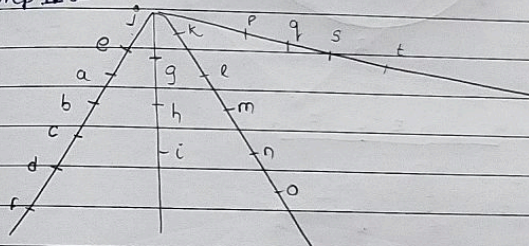
Step II :-



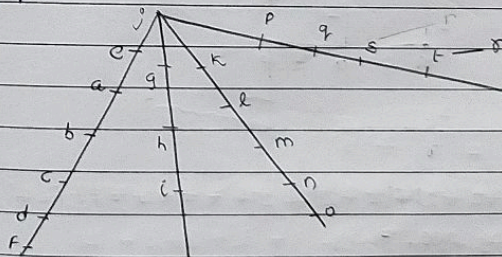
Step III



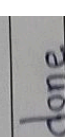
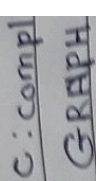
Step IV :-



Step V :-



load (no



Φ_{-27}

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